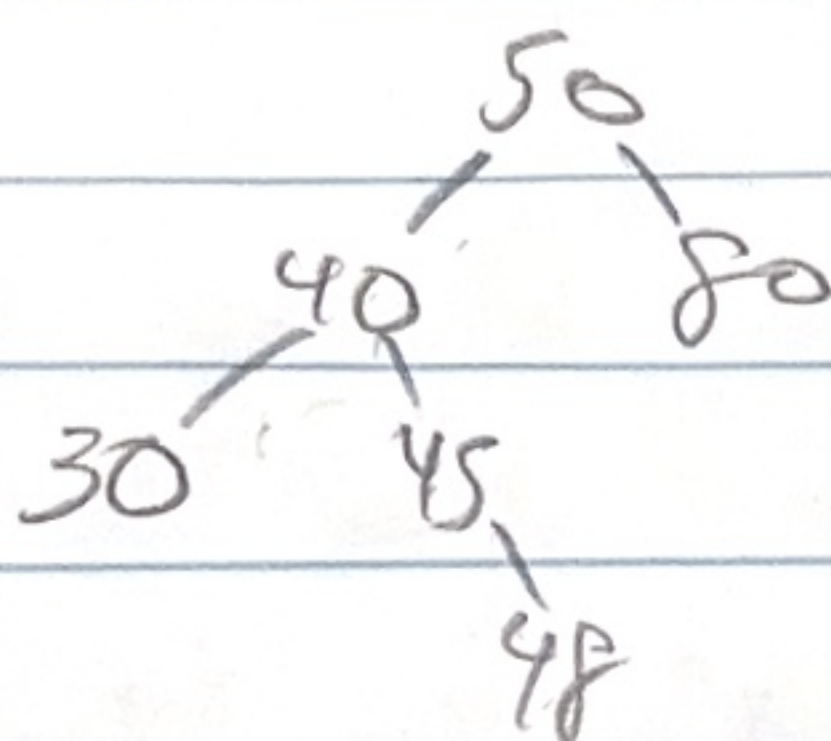
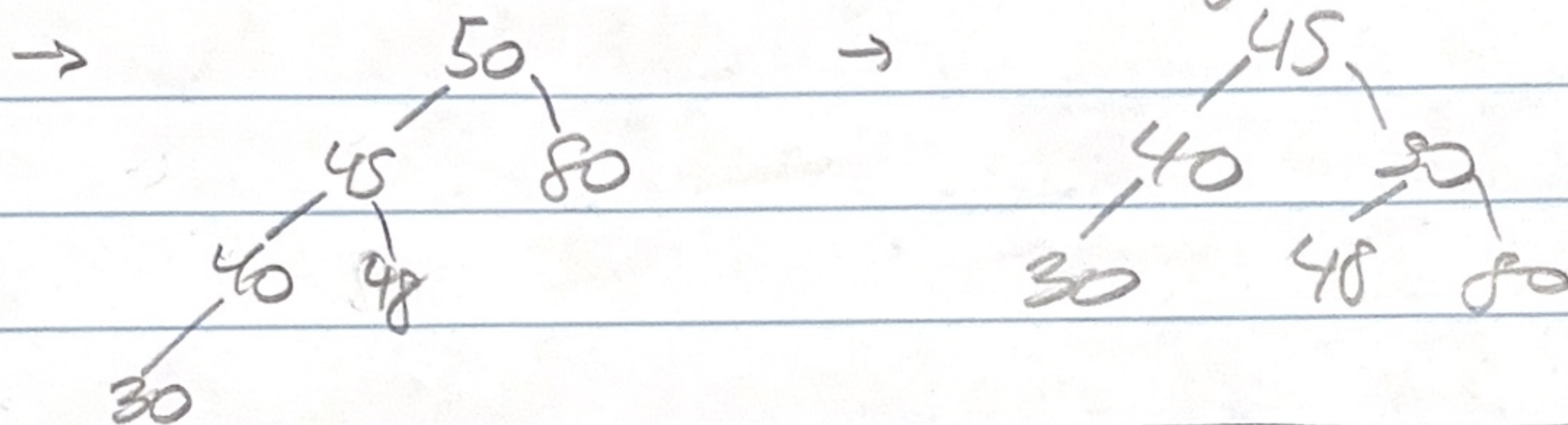


CSDS 233 #3 Q1

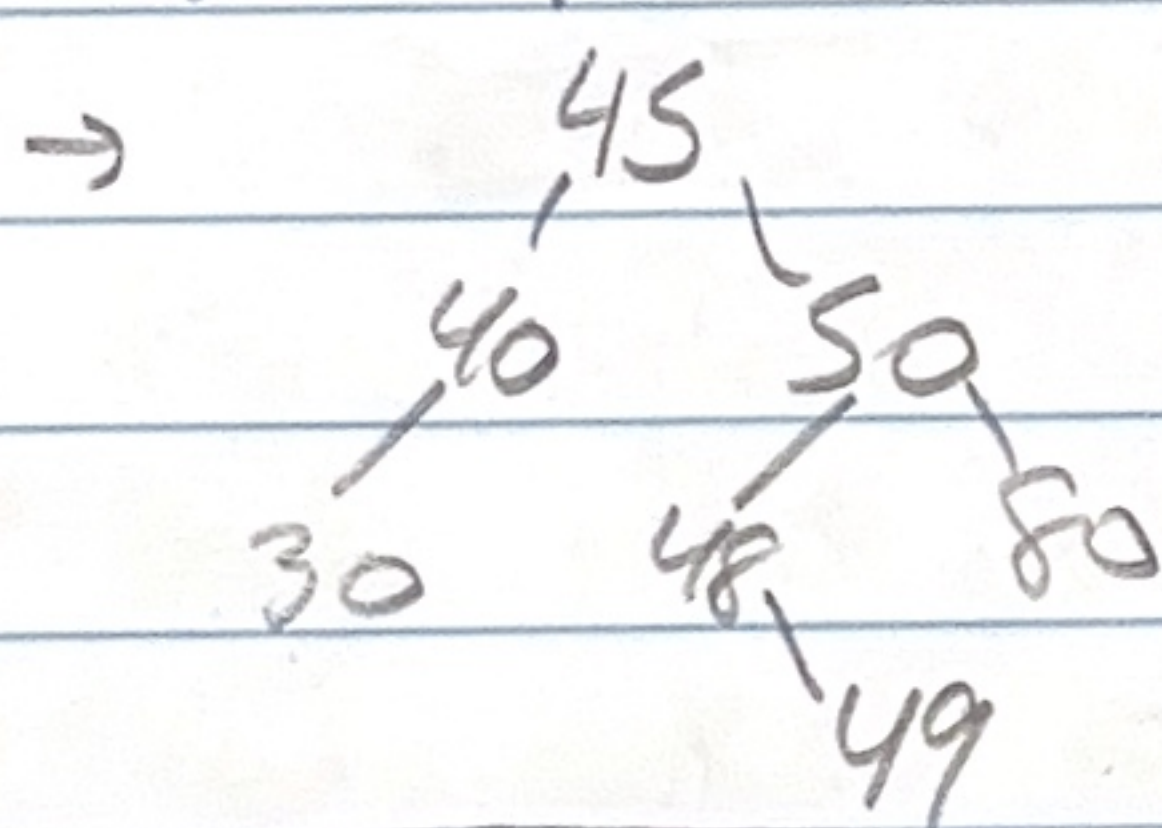
original tree:



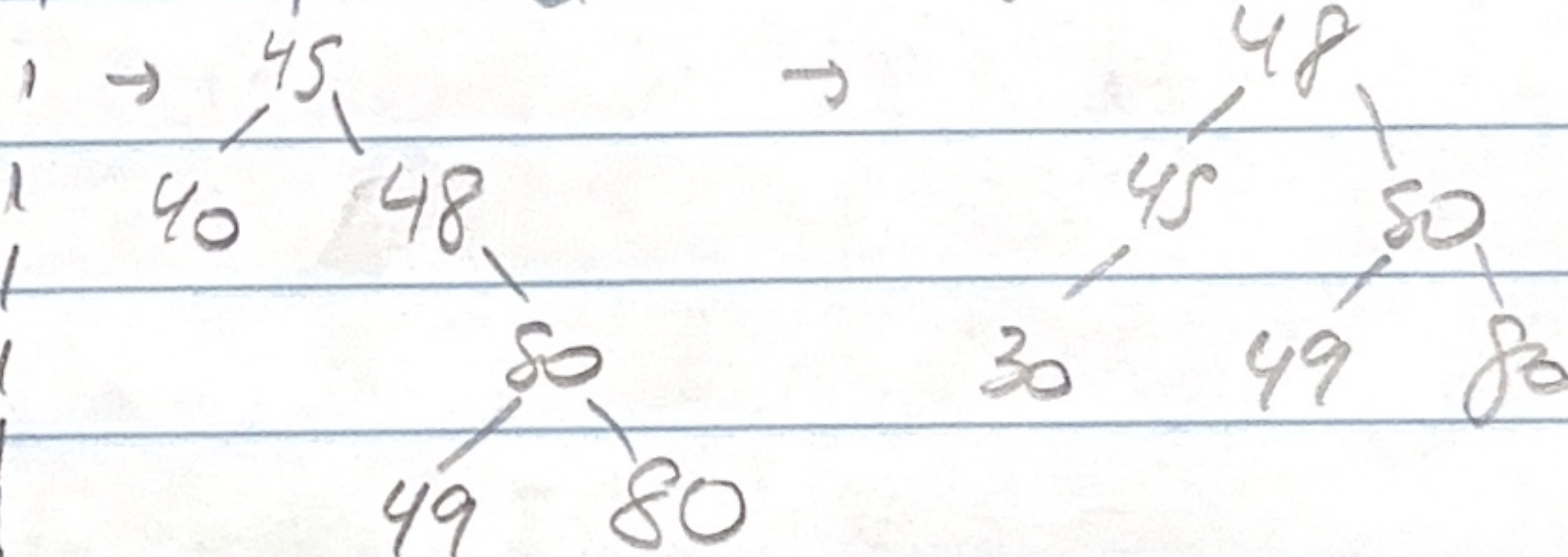
a) unbalance around 50 → left-right rotation



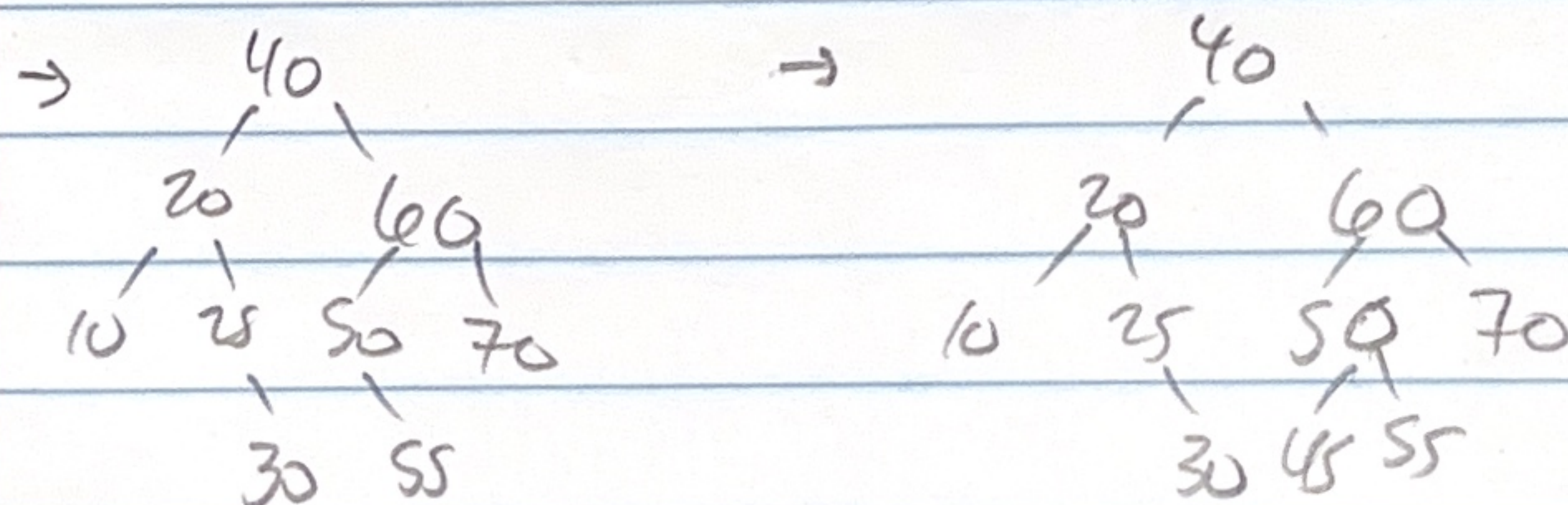
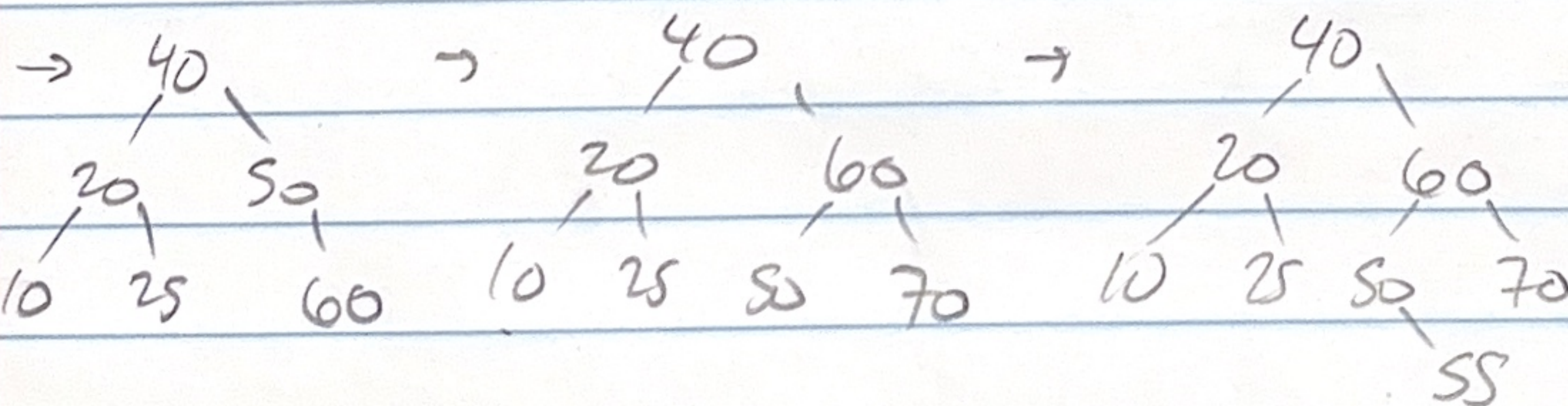
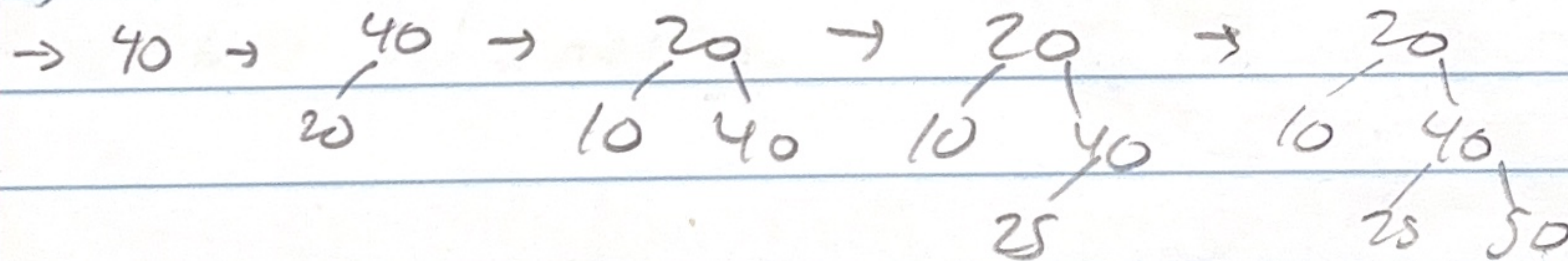
b) insert 49



c) delete 30 → right-left rotation



d) insert (in order) 40, 20, 10, 25, 50, 60, 70, 55, 30, 45



CSDS 233 #3 Q2

- a) False, a node must have at least $\frac{m}{2}$ nodes, but can have up to m .
- b) False, a full node will send its middle key up to its parent and split, and said node will not always be the root.
- c) True, each level of the tree can hold up to another multiple of $m-1$. (root could have $m-1$, and the next level could have up to $(m-1)^2$)
- d) True, like all trees, searching through it will have an average case of $O(\log m)$ but if it is basically a linked list then it becomes $O(m)$. Both cases heavily depend on m and the number of levels.
- e) False, if it remains over the minimum number of keys, nothing needs to happen. If it falls below that threshold, it can try borrowing keys from its neighbors before merging.

CSDS 233#3 Q3

a) A binary tree is just a tree where the maximum child count of any node is 2. A binary search tree is a binary tree that follows these rules: left child is less than the root/parent and right child is greater than the root/parent.

b) 12 nodes: to get a height of 4 you only need one node in the 4th level, then start deleting nodes from level 3 without making rotations or unbalancing the tree. There are 4 deletions you can make and $16 - 4 = 12$.

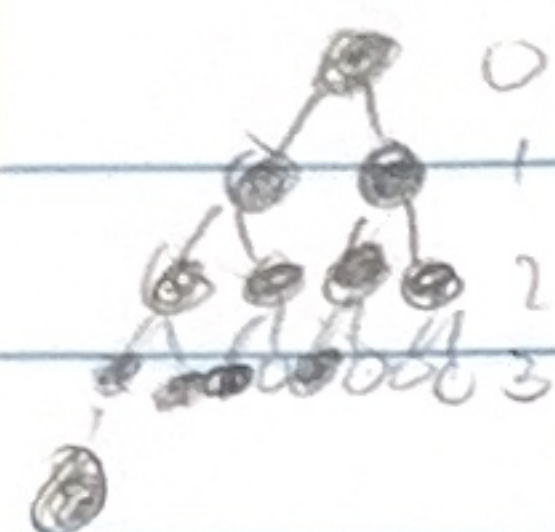
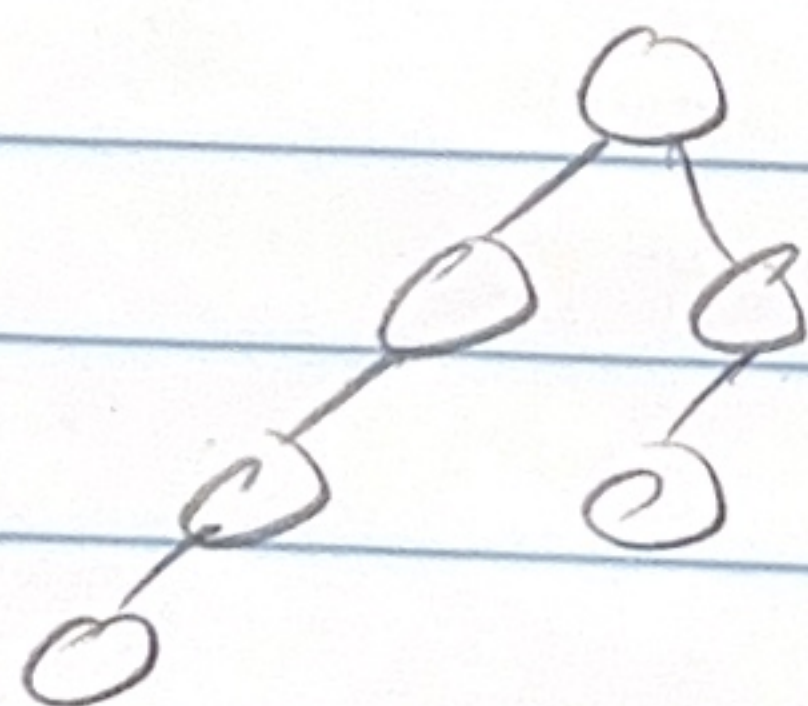
c) An AVL tree is a BST with a balance factor. Each node in the tree has a balance factor, but it is unimportant for leaf nodes, it is defined as the height of the right subtree - the height of the left subtree.

An imbalance occurs when one side of the tree is too much longer/heavier than the other, this is bad because it can impact tree operation efficiency.

Imbalances are mitigated by rotations that shift the order of the nodes.

d) A Balanced Binary Tree is a tree where the difference in height of the left and right subtree is no more than 1, an AVL tree has a balance factor at each node.

e) Balanced Tree, but not AVL



Ashar Chohan (axc1541)

WDS 233 #3 Q4

a) True

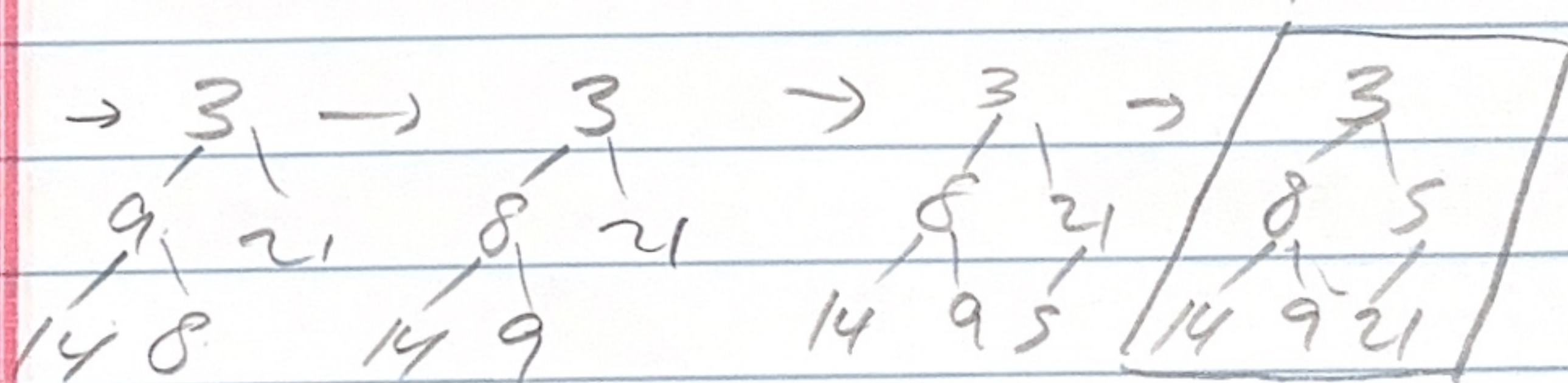
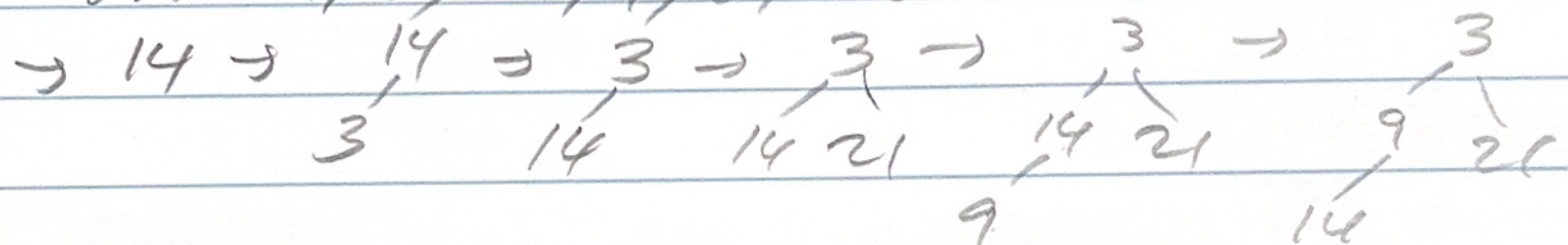
b) True

c) True

d) False, its $\log(n)$ because it traverses the height of the heap

e) True

Insert $[14, 3, 21, 9, 8, 5]$



Delete 3

